LogicDirector: Enforcing Temporal Composition in Text-to-Video Generation

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Abstract

Accurately rendering temporal composition is fundamental to text-to-video (T2V) generation. While recent efforts incorporate temporal structure for long-horizon coherence, the intrinsic temporal understanding capability of T2V models remains underexplored. We observe that current models still struggle to respect canonical operators such as *before* and *after*, usually accompanied by missing entities and incorrect action binding. In this work, we propose **LogicDirector**, a test-time guidance framework that enforces temporal composition through executable logical specifications. By translating text prompts into first-order logic constraints over attention-derived evidence, LogicDirector constructs a neuro-symbolic verifier to evaluate noisy latent states during diffusion sampling for entity grounding, action binding, and temporal adherence. To enforce these constraints without expensive gradient-based optimization, we introduce a gradient-free Best-of-n latent search strategy, which preserves the model's native diffusion dynamics while steering logical satisfaction. To systematically study operator-level temporal composition, we curate **TempBench**, a diagnostic benchmark spanning four canonical relations, together with a hierarchical evaluation protocol that disentangles entity existence, event realization, and temporal correctness. Experiments on CogVideoX and Wan 2.1 demonstrate that our method significantly improves temporal composition while maintaining visual fidelity. Code will be released at our project page.

1 Introduction

Text-to-video (T2V) generation [4, 17, 57, 60, 41, 52] has emerged as a key frontier in generative AI. Recent large-scale T2V systems [39, 10, 49] demonstrate impressive capabilities in synthesizing photorealistic and temporally coherent video clips. Despite this rapid progress, current models still struggle to accurately follow complex, compositional prompts, especially when the task requires reasoning about temporal composition, i.e., the ability to sequentially ground entities, accurately bind their actions, and enforce strict chronological relations between multiple events.

Recent efforts have begun to incorporate temporal structure into T2V generation [28, 50, 36], but they are typically motivated by long-horizon coherence: generating longer narratives by decomposing prompts into scenes or events and connecting them through scripts [21, 32], plans [27, 13, 22], or time-binding [58, 53]. This line of work improves multi-event coverage and sequential transition, yet it does not directly answer a fundamental question: *Do current T2V models intrinsically understand temporal composition at the object-event level?* In other words, even in short prompts with only two events, can the model reliably render the prerequisite actions and enforce temporal relations such as *before* and *after*?

In practice, we observe that the answer is often negative. As shown in Figure 1, state-of-the-art models may render both events but in the wrong order, collapse them into simultaneous motions,

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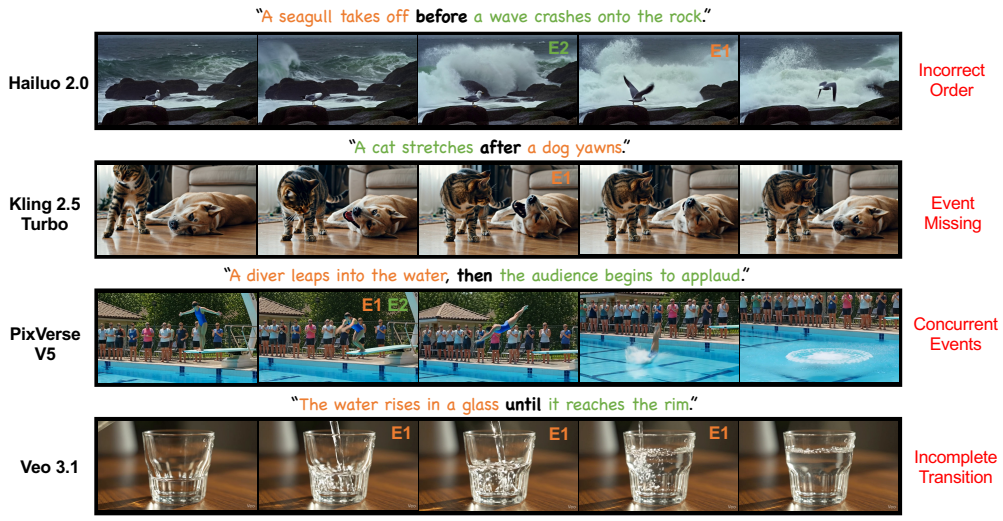


Figure 1: State-of-the-art T2V models still struggle with intrinsic temporal composition. Even when given simple, two-event compositional prompts containing temporal operators, models often fail by reversing the chronological order or dropping events entirely.

or take a trivial shortcut where one event never visibly occurs. Crucially, temporal violations often co-occur with missing entities or incorrect action binding, indicating that enforcing temporal relations requires reliable grounding of events and their participants as prerequisites. While some recent attempts [62, 7] build post-hoc verification or editing pipelines to correct temporal inconsistencies, they typically rely on external modules and iterative generate-and-fix loops rather than addressing the model's inherent limitations. This motivates our core research question: *Can we natively enforce explicit temporal composition during the generation process?*

To answer this question, we propose **LogicDirector**, a test-time guidance framework that embeds predicate logic directly within the denoising steps. Our key insight is to treat temporal composition not merely as narrative structures, but as executable specifications that can be evaluated and enforced during diffusion sampling. Specifically, we translate each prompt into a set of first-order propositions that describe entities, actions, and their temporal relations. Instead of operating in a discrete symbolic space, we relax these propositions into continuous objectives using fuzzy logic and apply them to the model's spatiotemporal cross-attention maps. These maps provide a natural interface between text tokens and visual tokens, allowing us to measure whether events are grounded, correctly bound to entities, and temporally ordered as intended.

Given these differentiable logical objectives, an intuitive idea is to directly optimize the noisy latents via gradient descent. However, this faces two practical limitations. First, backpropagating gradients through a large T2V backbone incurs substantial memory overhead, making it prohibitively expensive in practical inference settings. Second, aggressively minimizing the logic objective can easily distort the sampling trajectory, pushing the latent away from the learned manifold and severely degrading visual quality. To circumvent this, we propose a gradient-free Best-of-n latent search strategy. Rather than forcing the latent along an explicit gradient direction, we generate multiple locally perturbed candidates at each denoising step and evaluate them using the logic verifier. The candidate that best satisfies the logical constraints is selected to continue the denoising process. This strategy preserves the model's natural sampling dynamics while steering generation toward temporally compliant solutions, achieving explicit temporal control without memory-intensive backpropagation.

To systematically validate our method, we curate a diagnostic benchmark, **TempBench**, designed to evaluate operator-level temporal composition in T2V generation. Unlike broad evaluation suites [23, 31, 47] that emphasize visual fidelity, attribute binding, or coarse video-text alignment, TempBench comprises 200 prompts spanning four canonical relations (*before/after/until/then*), focusing on chronological order and termination. Extensive experiments demonstrate that LogicDirector consistently improves entity grounding, action binding, and temporal-relation adherence on two open-source backbones, CogVideoX [60] and Wan 2.1 [52], while preserving visual fidelity.

In summary, our main contributions are as follows:

- We identify intrinsic temporal composition as a fundamental yet underexplored limitation of current T2V models, systematically analyzing their failure modes at the fine-grained object-event level.
- We propose LogicDirector, a test-time guidance framework that translates prompts into first-order logic constraints over cross-attention maps and enforces operator-level temporal composition via a gradient-free Best-of-n latent search during early sampling.
- We introduce TempBench, a compact diagnostic benchmark with a hierarchical evaluation protocol that decomposes performance into entity existence, event realization, and temporal correctness, enabling fine-grained analysis of temporal composition behavior.

2 Related Work

Temporal-grounded text-to-video generation. Beyond single-clip synthesis, recent work targets *sequential* or *multi-event* T2V generation, where prompts describe multiple events or scene transitions. Representative approaches decompose long descriptions into event- or shot-level prompts and generate clips under continuity constraints, including progressive prompting for compositional generation [50], multi-sentence diffusion pipelines [36], and story-level diffusion models emphasizing transitions across clips [6]. More recent systems study multi-shot narratives and storyboard-style generation with discrete cuts and consistent characters across shots [24, 2]. In parallel, several methods pursue more direct temporal control over *when* events occur, e.g., via temporally grounded representations for event timing [16, 58]. While these efforts improve long-horizon coherence and sequencing, they rarely provide operator-level guarantees for canonical temporal relations. In contrast, we enforce such temporal composition at inference time via a structured logic verifier and test-time search.

Test-time guidance for diffusion models. A growing line of work improves generation controllability through inference-time control of diffusion latents or noise while keeping pretrained diffusion models largely frozen [51, 12, 64, 40]. Classifier-free guidance (CFG) [20] combines conditional and unconditional score estimates to balance prompt fidelity and sample diversity. Beyond CFG, prior work guides diffusion sampling with auxiliary energies or gradients [29, 61, 45], or adapts conditioning across denoising steps for finer semantic control [56, 35]. Another line of work uses cross-attention as an internal interface between text tokens and visual content, enabling spatial composition [59, 63], attribute binding [42, 5, 46], and image editing [19, 33, 44]. These methods demonstrate the utility of cross-attention for token-level grounding, but typically exploit it through task-specific heuristic controls. In contrast, we lift cross-attention into predicate-level temporal evidence, compose entity, event, binding, and relation constraints through first-order fuzzy logic, and use the resulting structured satisfaction score as an intrinsic feedback signal for temporal composition.

3 LogicDirector

Given a pre-trained T2V model and a text prompt, our goal is to enforce the temporal composition described in the prompt without additional model fine-tuning. To this end, we introduce **LogicDirector**, a test-time guidance framework that treats temporal composition as an executable specification evaluated directly during diffusion sampling.

As illustrated in Figure 2, the main branch follows the standard denoising diffusion process, where a text prompt and a noisy latent $\mathbf{z}_t$ are fed into the video generation model to produce progressively refined predictions. In parallel, we use the GPT-4.1 API [37] to parse the input prompt into a structured specification consisting of required entities, actions, and a temporal operator. This specification is converted into first-order logic objectives defined over the model's spatiotemporal cross-attention maps. Rather than optimizing the latent through gradient descent, we incorporate these objectives via a gradient-free Best-of-n latent search strategy. At each denoising step, we sample multiple locally perturbed candidates $\{\tilde{\mathbf{z}}_t^{(i)}\}_{i=1}^{n}$, evaluate their logical losses via a batched forward pass, and select the best candidate to continue the trajectory. This selection-based integration guides denoising toward trajectories that better satisfy the specification, while preserving the native diffusion dynamics and computational efficiency.

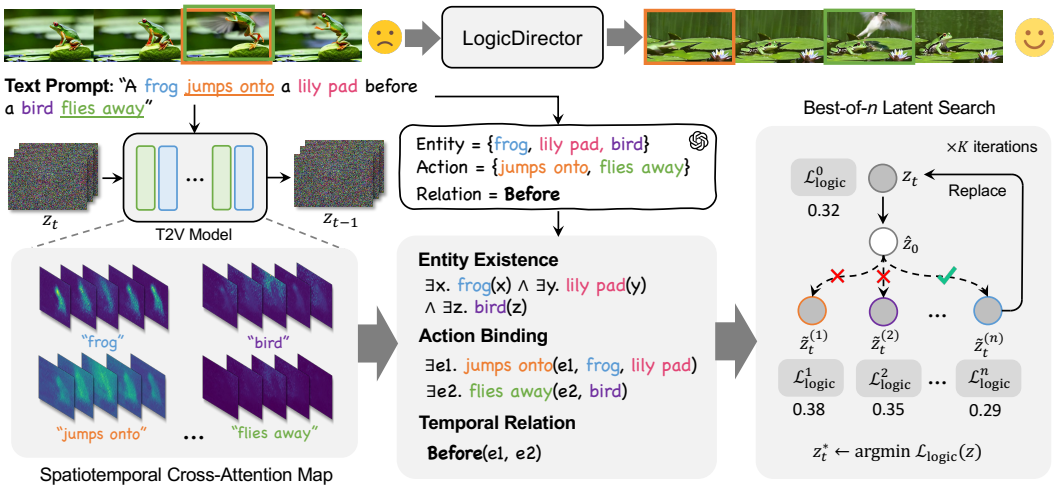


Figure 2: An overview of the **LogicDirector** framework. It decomposes a text prompt into entities, actions, and relations, formulates them as first-order logic constraints, and evaluates using spatiotemporal cross-attention evidence during denoising. A gradient-free Best-of-n latent search then selects the most logically consistent candidate to steer sampling toward temporally compliant videos.

3.1 Preliminaries

To achieve temporal composition, we first translate a natural language prompt into a structured logical specification that captures entities, actions, and their temporal relations. We adopt first-order logic (FOL) [43, 11] as the representational framework for modeling objects and their interactions. In FOL, semantics are expressed through: (i) **Variables**, which represent entities or events (e.g., x for a frog, y for a lily pad); (ii) **Predicates**, which describe properties of objects or relations between them (e.g., $\texttt{JumpOnto}(x, y)$, $\texttt{FlyAway}(z)$); and (iii) **Quantifiers**, such as $\exists$ ("there exists") and $\forall$ ("for all"), which specify the scope of these variables. To capture temporal relations, we introduce a domain-specific set of temporal operators—`BEFORE`, `AFTER`, `THEN`, and `UNTIL`—which are defined as predicates over events [8]. For example, given the prompt "a frog jumps onto a lily pad before a bird flies away," we construct the logical specification:

$$\exists x, y, z, e_1, e_2 \ \texttt{Frog}(x) \wedge \texttt{LilyPad}(y) \wedge \texttt{Bird}(z) \\ \wedge \texttt{JumpOnto}(e_1, x, y) \wedge \texttt{FlyAway}(e_2, z) \wedge \texttt{BEFORE}(e_1, e_2).$$

To connect the discrete logical specification with neural generation, we relax each predicate into a continuous score derived from the model's spatiotemporal cross-attention maps. Let $\mathcal{C}$ denote the set of logical predicates parsed from the prompt specification, including entity predicates and action predicates. Each predicate $c \in \mathcal{C}$ corresponds to one or more tokens in the prompt. During denoising, we extract cross-attention maps between video tokens and the corresponding text tokens from the diffusion backbone and aggregate them across attention heads and layers. This yields an attention map $A_c \in [0, 1]^{T \times H \times W}$ for each predicate c, where $A_c[t, h, w]$ indicates how strongly the visual token at (t, h, w) attends to the text tokens corresponding to c, grounding logical predicates in the model's internal attention activations. We further adopt product fuzzy logic [18] to relax Boolean operators into continuous functions, as shown in Table 1, which lays the foundation for a structured verification and enforcement of temporal composition.

3.2 Logical Constraints over Cross-Attention

We now describe how the logical specification introduced in Sec. 3.1 is instantiated as continuous constraint functions defined over spatiotemporal cross-attention. Following the structure of temporal composition, we first ensure that entities and actions are properly grounded and correctly bound, and then enforce temporal relations between events.

Entity and Action Existence. To measure whether a predicate c is visually grounded, we aggregate spatial activations of its cross-attention map using top-k mean to obtain a frame-level activation score:

$$\alpha_c(t) = \text{TopKMean}(A_c[t, :, :]) \,. \tag{1}$$

Table 1: Fuzzy logic operators applied to spatio-temporal cross-attention maps. A_P, A_Q denote the attention maps associated with predicates P and Q. The index i refers to a spatio-temporal location (t, h, w), and $\odot$ denotes the Hadamard product.

Operator	**Linguistic Meaning**	**Differentiable Formulation**
$P(x)$	Predicate truth	$A_P[i]$
$\neg P(x)$	Negation	$1 - A_P[i]$
$P(x) \wedge Q(x)$	Conjunction	$A_P[i] \odot A_Q[i]$
$P(x) \vee Q(x)$	Disjunction	$1 - (1 - A_P[i]) \odot (1 - A_Q[i])$
$P(x) \Rightarrow Q(x)$	Implication	$1 - A_P[i] \odot (1 - A_Q[i])$
$\forall x.\, P(x)$	Universal quantifier	$\prod_i A_P[i]$
$\exists x.\, P(x)$	Existential quantifier	$1 - \prod_i (1 - A_P[i])$

We then calibrate $\alpha_c(t)$ into a soft presence probability $p_c(t) = \sigma((\alpha_c(t) - \gamma)/\tau)$, where γ is an activation threshold and τ denotes a scaling factor. Finally, we define the overall existence score as

$$\mathcal{L}_{exist}(c) = -\log\Big(1 - \prod_t (1 - p_c(t))\Big). \tag{2}$$

This objective encourages at least one *latent* frame to strongly activate predicate c while remaining robust to noisy attention. The logical specification determines which predicates must be grounded in the video, while the attention-based scores provide a soft visual proxy to verify their existence.

Action-Entity Binding. Beyond existence, action predicates must be correctly bound to their participating entity predicates. Given an action predicate $a \in \mathcal{C}$, let $\mathcal{E}(a) \subset \mathcal{C}$ denote the set of entity predicates associated with a in the prompt specification. We enforce binding through a fuzzy implication that encourages entity co-activation wherever the action activates:

$$\mathcal{L}_{bind}(a) = -\frac{1}{|\mathcal{E}(a)|} \sum_{e \in \mathcal{E}(a)} \mathbb{E}_{t,h,w}\Big[\log\big(1 - A_a[t,h,w]\big(1 - A_e[t,h,w]\big)\big)\Big]. \tag{3}$$

This constraint encourages that whenever an action predicate becomes active, its required entity predicates co-activate at the same spatiotemporal locations, reducing action misbinding and providing a reliable foundation for subsequent temporal composition.

Temporal Operators. Having obtained frame-level activation scores $\alpha_c(t)$ for each predicate, we define temporal operators that regulate their ordering.

`BEFORE / AFTER` – The operator `BEFORE`(p, q) requires that predicate p occur prior to predicate q. Given activation scores $\alpha_p(t)$ and $\alpha_q(t)$, we first encourage the existence of at least one correctly ordered frame pair:

$$S_{before}(p, q) = 1 - \prod_{t_i < t_j} \big(1 - \alpha_p(t_i)\alpha_q(t_j)\big). \tag{4}$$

This term acts as a soft existential condition over ordered frame pairs (t_i, t_j) with $t_i < t_j$, encouraging activation of p to appear earlier than that of q. To prevent temporal inversion or excessive overlap, we additionally penalize activations in the reverse region:

$$S_{rev}(p, q) = 1 - \prod_{t_j \le t_i} \big(1 - \alpha_p(t_i)\alpha_q(t_j)\big). \tag{5}$$

Therefore, the final loss is defined as:

$$\mathcal{L}_{before}(p, q) = -\log S_{before} - \log\big(1 - S_{rev}\big). \tag{6}$$

The `AFTER`(p, q) operator is defined symmetrically as $\mathcal{L}_{after}(p, q) = \mathcal{L}_{before}(q, p)$.

`THEN` – While *before* only enforces onset ordering, *then* typically requires non-overlapping succession, i.e., q should begin only after p concludes. Accordingly, `THEN`(p, q) restricts valid frame pairs to those separated by at least Δ frames:

$$S_{then}(p, q) = 1 - \prod_{t_j \ge t_i + \Delta} \big(1 - \alpha_p(t_i)\alpha_q(t_j)\big). \tag{7}$$

To discourage premature or overlapping activation, we retain the reverse-order penalty $S_{rev}(p, q)$ defined above. The resulting loss is

$$\mathcal{L}_{then}(p, q) = -\log S_{then} - \log\big(1 - S_{rev}\big). \tag{8}$$

`UNTIL` – The operator $\texttt{UNTIL}(p, q)$ models a state-maintaining relation in which predicate p persists until predicate q occurs. We define the cumulative occurrence score of q as $S_{\leq t}(q) = 1 - \prod_{\tau \leq t}\big(1 - \alpha_q(\tau)\big)$, which represents the probability that q has occurred by frame t. The objective encourages p to remain active before q occurs and to cease afterward:

$$\mathcal{L}_{until}(p, q) = -\mathbb{E}_t\Big[(1 - S_{\leq t}(q))\, \log \alpha_p(t) + S_{\leq t}(q)\, \log\big(1 - \alpha_p(t)\big)\Big]. \tag{9}$$

The cumulative score $S_{\leq t}(q)$ acts as a soft temporal gate: before q occurs, the first term penalizes missing activation of p; after q occurs, the second term penalizes continued activation of p. Existence constraints on q ensure that the triggering event eventually materializes.

Final Logic Objective. We aggregate the above constraints into a unified logic verifier that measures overall satisfaction of temporal composition. Given a prompt specification with predicate set $\mathcal{C}$, the final loss is:

$$\mathcal{L}_{logic} = \lambda_1 \sum_{c \in \mathcal{C}} \mathcal{L}_{exist}(c) + \lambda_2 \sum_{a \in \mathcal{C}_{act}} \mathcal{L}_{bind}(a) + \lambda_3 \mathcal{L}_{rel}(p, q), \tag{10}$$

where $\mathcal{C}_{act} \subset \mathcal{C}$ denotes the set of action predicates, and $\mathcal{L}_{rel}$ is the operator-specific temporal loss (`BEFORE/AFTER/UNTIL/THEN`) defined on the event pair (p, q). This objective serves as a continuous logical verifier that evaluates whether a candidate generation trajectory satisfies the structured temporal specification derived from the prompt.

3.3 Best-of-n Latent Search

Having defined the logical objective, we now describe how to integrate it into diffusion sampling at inference time. A direct approach would be to optimize the noisy latent by backpropagating through the logic loss. However, this requires storing activations through a large T2V backbone and can be prohibitively memory-intensive during inference. Moreover, aggressive latent optimization may push the trajectory away from the learned generative manifold, leading to visual artifacts and degraded sample quality. We therefore use the logical objective as a *verifier* rather than a gradient signal, and perform a gradient-free Best-of-n local search in latent space.

As illustrated in Figure 2, at a denoising step t, we first obtain the model's one-step clean latent estimate $\hat{\mathbf{z}}_0$ from the current noisy latent $\mathbf{z}_t$ using the backbone's native prediction parameterization (see Appendix B for details). We then re-inject noise using the backbone's noise schedule at the same diffusion level to construct a set of local stochastic proposals:

$$\tilde{\mathbf{z}}_t^{(i)} = a_t \hat{\mathbf{z}}_0 + b_t \boldsymbol{\epsilon}^{(i)}, \qquad \boldsymbol{\epsilon}^{(i)} \sim \mathcal{N}(0, \mathbf{I}), \quad i = 1, \ldots, n. \tag{11}$$

Here, a_t and b_t are the coefficients of the clean latent and Gaussian noise, respectively, as determined by the backbone's forward noising process. Their model-specific values are given in Appendix B. Together with the original latent $\mathbf{z}_t$, these proposals form a localized candidate set around the current denoising trajectory. Branching from the shared estimate $\hat{\mathbf{z}}_0$ allows exploration of nearby feasible trajectories while remaining compatible with the diffusion prior.

We then evaluate each candidate by running a forward pass to obtain its cross-attention maps and computing the corresponding logical loss. Since the verifier is defined directly over attention-derived predicates, this evaluation requires no external reward model or additional training. Let $\mathcal{Z}_t = \{\mathbf{z}_t, \tilde{\mathbf{z}}_t^{(1)}, \ldots, \tilde{\mathbf{z}}_t^{(n)}\}$ denote the candidate set. We select the candidate latent with the lowest logical loss:

$$\mathbf{z}_t^* \leftarrow \arg\min_{\mathbf{z} \in \mathcal{Z}_t} \mathcal{L}_{logic}(\mathbf{z}). \tag{12}$$

This selection step can be repeated for K inner iterations at each timestep. Since each round proposes new candidates around the current best latent, the procedure can be viewed as stochastic hill-climbing under the logic verifier. Applied during early denoising stages, this local search biases the trajectory toward higher logical satisfaction while preserving the backbone's native sampling dynamics.

Table 2: Comparison with existing T2V benchmarks. Our TempBench offers balanced coverage of diverse temporal relations.

Benchmark	*Before*	*After*	*Until*	*Then*	Total
VBench [23]	✗	✓	✗	✗	1
EvalCrafter [31]	✓	✓	✗	✗	3
TC-Bench [14]	✗	✓	✗	✓	3
FETV [30]	✓	✓	✗	✓	72
StoryEval [55]	✓	✓	✗	✓	409
T2V-CompBench [47]	✓	✓	✓	✓	12
TempBench (Ours)	✓	✓	✓	✓	**200**

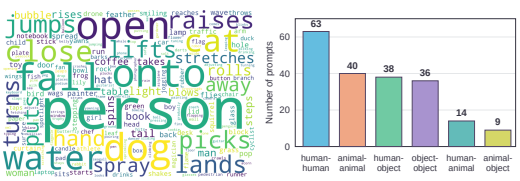


Figure 3: Prompt suite statistics. (*Left*) Word cloud illustrating the vocabulary distribution of our TempBench prompt suite. (*Right*) Number of prompts across different evaluation categories.

4 Experiments

4.1 Experimental Setup

Benchmark. To systematically evaluate operator-level temporal composition, we first examine the coverage of existing T2V benchmarks. As shown in Table 2, most benchmarks either omit canonical temporal relations or provide insufficient coverage for systematic analysis. To this end, we curate TempBench, a compact diagnostic benchmark covering four canonical operators: *before*, *after*, *until*, and *then*. TempBench consists of 200 carefully constructed prompts, designed to feature visually distinguishable event boundaries and temporally concise sequences. As illustrated in Figure 3, the resulting suite spans common entities such as humans, animals, and everyday objects, thereby evaluating a broad spectrum of actions and interactions.

Evaluation Metrics. We employ a three-stage evaluation framework using Qwen3-VL-32B Thinking [3] as a structured visual verifier. The protocol follows a strict dependency chain: (i) entity existence (S_{ent}), computed as the fraction of required entities visually present in each video; (ii) event realization (S_{evt}), which measures the average success rate of executing the required dynamic actions (two events per prompt); and (iii) temporal relation satisfaction (S_{rel}), which reports the strict success rate of the target temporal operator. Crucially, S_{rel} is evaluated *if and only if* all required events are visually confirmed, preventing inflated scores caused by incomplete generations. This hierarchical design disentangles grounding failures from genuine temporal violations, enabling fine-grained analysis of compositional reasoning. More details are provided in Appendix A.

Implementation Details. We evaluate LogicDirector on two representative open-source T2V backbones: CogVideoX-2B [60] and Wan 2.1-1.3B [52]. For logical constraint computation, we set the activation threshold γ to 0.04 for entities and 0.025 for actions in CogVideoX, and to 0.025 for both entities and actions in Wan 2.1. The sigmoid scaling factor τ is fixed at 0.01 for both models. For the `THEN` operator, the non-overlapping interval Δ is set to 5 for CogVideoX and 3 for Wan 2.1. For Best-of-n latent search, we set proposal candidate $n = 1$ and perform $K = 3$ iterations per timestep during the first 10 denoising steps. All experiments are conducted on six NVIDIA RTX 6000 Blackwell GPUs. More implementation details can be found in Appendix B.

4.2 Main Results

In this section, we conduct a systematic evaluation of 10 text-to-video generation models on TempBench. As shown in Table 3, we observe that: (1) Closed-source models achieve near-saturated entity grounding and high event realization, yet their temporal success remains moderate. Even the strongest model, Hailuo 2.0 [34], reaches at most $\sim$60% average S_{rel}. This suggests that temporal operators cannot be reduced to generic video-text alignment. (2) Compared to commercial models, open-source baselines perform substantially worse, and the bottleneck varies across backbones. Even the strongest open model, Wan2.1-14B, reaches only 14.0% average S_{rel}. Notably, *Then* is the easiest operator for open models, whereas *Before* and especially *After* remain challenging. This suggests that non-overlapping succession is easier for current generators than fine-grained onset ordering. (3) Our LogicDirector consistently improves temporal correctness on both evaluated backbones, nearly doubling the S_{rel} on the most challenging operators. While entity grounding also improves, the strongest gains are in event realization and relation adherence, suggesting that LogicDirector primarily strengthens compositional reasoning rather than merely boosting entity-token presence.

Table 3: TempBench evaluation results (%) on 10 text-to-video generation models. The best results among open-source models are in **bold** and the second best is underlined.

Models	*Before*			*After*			*Until*			*Then*			Avg.		
	S_{ent}	S_{evt}	S_{rel}	S_{ent}	S_{evt}	S_{rel}	S_{ent}	S_{evt}	S_{rel}	S_{ent}	S_{evt}	S_{rel}	S_{ent}	S_{evt}	S_{rel}
(Closed-Source)															
Pika2.2 [26]	96.0	75.0	48.0	96.3	74.0	44.0	96.7	73.0	44.0	96.5	79.0	58.0	96.4	75.3	48.5
Kling 2.5-Turbo [48]	94.7	72.0	46.0	94.0	65.0	24.0	93.0	68.0	34.0	98.3	68.0	40.0	95.0	68.3	36.0
Hailuo 2.0 [34]	96.7	80.0	52.0	96.7	82.0	56.0	95.7	80.0	62.0	92.8	88.0	66.0	95.5	82.5	59.0
PixVerse V5 [1]	99.3	87.0	68.0	96.7	73.0	36.0	98.0	79.0	48.0	98.7	83.0	60.0	98.2	80.5	53.0
(Open-Source)															
Mochi [15]	83.7	42.0	8.0	82.7	52.0	14.0	85.3	47.0	18.0	82.0	52.0	26.0	83.4	48.3	16.5
CogVideoX-5B [60]	76.9	39.0	6.0	72.3	35.0	8.0	76.3	37.0	10.0	83.0	38.0	8.0	77.1	37.3	8.0
Wan2.1-14B [52]	**88.8**	46.0	10.0	**92.8**	42.0	6.0	**93.0**	47.0	14.0	**97.2**	62.0	26.0	**93.0**	49.3	14.0
HunyuanVideo [25]	88.3	46.0	14.0	92.0	**54.0**	6.0	87.0	38.0	8.0	93.2	53.0	28.0	90.1	47.8	14.0
CogVideoX-2B [60]	82.6	45.0	6.0	79.9	44.0	12.0	76.0	35.0	6.0	85.0	54.0	18.0	80.9	44.5	10.5
+ LogicDirector (Ours)	88.1	**47.0**	12.0	84.5	53.0	**18.0**	85.3	48.0	16.0	93.2	55.0	24.0	87.8	50.8	17.5
Wan2.1-1.3B [52]	86.2	45.0	12.0	86.1	38.0	6.0	90.3	45.0	12.0	86.3	58.0	22.0	87.2	46.5	13.0
+ LogicDirector (Ours)	85.3	46.0	**24.0**	90.8	45.0	12.0	90.3	**58.0**	**26.0**	94.2	**63.0**	**36.0**	90.2	**53.0**	**24.5**

Table 4: Ablation study of LogicDirector constraints on CogVideoX-2B.

Logic Constraints			*Before*			*After*			*Until*			*Then*			Avg.		
$\mathcal{L}_{exist}$	$\mathcal{L}_{bind}$	$\mathcal{L}_{rel}$	S_{ent}	S_{evt}	S_{rel}	S_{ent}	S_{evt}	S_{rel}	S_{ent}	S_{evt}	S_{rel}	S_{ent}	S_{evt}	S_{rel}	S_{ent}	S_{evt}	S_{rel}
✗	✗	✗	82.6	45.0	6.0	79.9	44.0	12.0	76.0	35.0	6.0	85.0	54.0	18.0	80.9	44.5	10.5
✓	✗	✗	86.2	42.0	8.0	85.5	43.0	10.0	82.0	34.0	6.0	83.2	52.0	18.0	84.2	42.8	10.5
✓	✓	✗	83.2	47.0	10.0	82.0	47.0	12.0	80.0	40.0	8.0	87.2	57.0	16.0	83.1	47.8	11.5
✓	✓	✓	88.1	47.0	12.0	84.5	53.0	18.0	85.3	48.0	16.0	93.2	55.0	24.0	87.8	50.8	17.5

4.3 Ablation Study

Contribution of each logical constraint. As shown in Table 4, the $\mathcal{L}_{exist}$ constraint consistently improves entity grounding over the baseline but yields minimal gains in temporal correctness. Moreover, S_{evt} slightly decreases, suggesting that enforcing token-level presence alone may encourage the model to render static entities without realizing the required motion. Notably, adding $\mathcal{L}_{bind}$ on top of $\mathcal{L}_{exist}$ reverses this trend, which increases S_{evt} from 42.8% to 47.8% on average, indicating that spatial co-activation between action and entity tokens helps the model produce verifiable events. Finally, enabling the temporal relation constraint $\mathcal{L}_{rel}$ yields the most consistent gains, improving S_{rel} by 6.0% on average while further boosting entity grounding and event realization. The effect is especially pronounced for *Until* and *After*, where correct triggering and ordering are difficult to obtain from grounding alone. Overall, the full verifier provides complementary signals that jointly reduce missing-content failures and temporal violations.

Latent Refinement Setups. We ablate three core hyperparameters of our LogicDirector guidance: the candidate set size n, inner-loop iterations K, and the number of guided denoising steps T. As shown in Table 5, we notice that: (1) Increasing K yields the most reliable gains: K=3 substantially improves temporal correctness over K=1, indicating that iterative local refinement is more effective than a single-step update. (2) Increasing n offers limited additional benefit but significantly increases runtime, suggesting the verifier benefits more from stable refinement than wider stochastic exploration. (3) Overly aggressive refinement leads to diminishing or negative returns, as late-stage perturbations can disrupt formed details and over-optimization may induce shortcut solutions. Qualitative comparisons can be found in Appendix C.

Table 5: Comparison with different hyperparameter setups on CogVideoX-2B.

n	K	T	S_{ent}	S_{evt}	S_{rel}	Time (s)
1	1	10	85.9	46.5	11.5	52
1	3	10	87.8	50.8	17.5	64
3	3	10	86.4	48.3	16.5	95
1	5	10	86.0	47.8	14.0	76
1	3	20	87.1	49.3	16.5	79



Figure 4: Qualitative comparison between baseline results and LogicDirector.

Evaluation on VBench. A natural concern is whether enforcing temporal composition during sampling degrades visual fidelity. To examine this, we utilize VBench [23] to evaluate the visual quality of our generated videos. Interestingly, LogicDirector does not degrade visual fidelity; instead, it consistently maintains or improves several metrics. As shown in Table 6, we observe gains in overall consistency, subject consistency, and image quality on CogVideoX-2B, and similar trends hold for Wan2.1-1.3B. We attribute these gains primarily to the Best-of-n latent search: among locally re-noised candidates at the same diffusion level, some trajectories exhibit fewer artifacts and better cross-frame stability, and logic-based selection implicitly filters for these higher-quality samples. Notably, the video dynamics show a mild decrease in some cases, which we attribute to suppressing temporally inconsistent or abrupt motions that tend to violate the imposed logical constraints.

Table 6: Quantitative evaluation on VBench.

Models	Overall Consist.	Subject Consist.	Background Consist.	Aesthetic Quality	Image Quality	Dynamic Degree	Motion Smooth.
CogVideoX-2B	23.13	89.23	93.91	47.25	53.95	73.00	97.87
+ LogicDirector (Ours)	23.90	91.43	94.50	50.11	60.74	68.50	97.88
Wan2.1-1.3B	23.82	87.42	92.46	49.46	59.68	84.50	97.98
+ LogicDirector (Ours)	24.25	87.50	92.42	48.90	60.56	83.50	98.30

4.4 Qualitative Analysis

Finally, we provide qualitative comparisons to further illustrate the effectiveness of LogicDirector. Figure 4 presents representative examples between baseline models and LogicDirector. Overall, LogicDirector reliably renders required entities and actions that are missed by baselines and enforces the intended sequence across different temporal compositions. We further compare our method with leading commercial models on a challenging *before* prompt. As shown in Figure 5, while closed-source models produce higher-fidelity videos, they frequently collapse the traffic light (e.g., *red* and *green*) into a co-occurring state and miss the specified ordering, whereas LogicDirector renders the two events separately and in the correct order, showing effective operator-level temporal control at inference time. More results can be found in Appendix E.



Figure 5: Qualitative comparison of LogicDirector with the leading commercial models.

5 Conclusion

In this work, we study intrinsic temporal composition in text-to-video generation. We find that current models often fail to satisfy compositional prompts, where temporal violations frequently co-occur with missing entities and event incompletion. To address this, we propose LogicDirector, a test-time guidance framework that enforces temporal composition via first-order logic constraints and steers sampling with a gradient-free Best-of-n latent search. Experiments on TempBench show that LogicDirector consistently improves entity grounding, event realization, and temporal correctness across two open-source backbones with only modest inference overhead. Looking ahead, while our current operators cover four canonical relations, we plan to expand to richer multi-event compositions, and to model finer temporal distance patterns, e.g., distinguishing immediate succession from long gaps, using stronger yet lightweight internal signals for grounding and temporal ordering.

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A TempBench

A.1 Prompt Suite Construction

TempBench is designed as a compact diagnostic benchmark for evaluating temporal composition in text-to-video generation. This section provides additional details on operator definitions and prompt construction.

Canonical Temporal Operators. TempBench focuses on four canonical temporal relations: *before*, *after*, *until*, and *then*. These operators capture common forms of temporal ordering and termination in short event sequences involving two events P and Q:

- *Before*: Start(P) $<$ Start(Q), i.e., P begins before Q. (e.g., "A person smiles *before* waving a hand.")
- *After*: Start(P) $>$ Start(Q), i.e., P begins after Q. (e.g., "A person smiles *after* waving a hand.")
- *Until*: P persists until onset of Q (e.g., "The sun sets *until* it is below the horizon.")
- *Then*: End(P) $\leq$ Start(Q), i.e., Event Q occurs only after P completes. (e.g., "A person closes a laptop, *then* puts it into a bag.")

Notably, we deliberately exclude relations such as *while* or *during*, which primarily capture co-occurrence rather than ordered or terminating event structure. Such relations often lack clear boundaries under short generated clips and are less suitable for strict temporal verification.

Prompt Generation and Curation. To construct a diverse yet verifiable prompt suite, we adopt a generate-curate pipeline. First, we oversample candidate prompts using two large language models, GPT-5 [38] and Gemini 2.5 Flash [9]. Each candidate prompt explicitly describes two events connected by one of the target temporal operators. The exact prompt template used for generation is provided in Figure 6. Second, all candidates undergo a manual curation pass to ensure that each prompt satisfies the following criteria:

- **Two-event structure:** The prompt must describe exactly two events with a clear temporal relation.
- **Visually distinguishable boundaries:** Both events must have clear visual onset and completion signals.
- **Temporally concise sequence:** The full event sequence should plausibly occur within a short generated clip.
- **Minimal semantic ambiguity:** Prompts requiring extensive commonsense inference or invisible states are removed.
- **Clear entity grounding:** All referenced entities should be visually identifiable.

After filtering, we obtain a balanced benchmark consisting of 200 prompts, with 50 prompts per operator.

Prompt Diversity: To ensure diverse coverage of real-world interactions, the prompt suite includes a broad range of entities (e.g., humans, animals, and everyday objects) and spans multiple action categories, as shown in Figure 3. Representative examples for each operator are shown in Table 7.

A.2 Verification Protocol

Evaluating temporal relations in generated videos requires verifying both event occurrence and event ordering. We therefore adopt a hierarchical three-stage verification framework using Qwen3-VL-32B Thinking [3] as a structured visual verifier. The verification pipeline consists of three stages: (i) Entity existence; (ii) Event existence; and (iii) Temporal relation. Notably, temporal relation is evaluated *if and only if* all required entities and events are visually confirmed, which prevents inflated temporal scores caused by incomplete event realizations.

Entity Existence. This dimension verifies that all required entities mentioned in the prompt are visually present in the generated video. The prompt template used for entity existence is provided

System-Msg: You are an expert AI researcher constructing a rigorous benchmark to evaluate Text-to-Video (T2V) models on their understanding of temporal composition. Your task is to generate short, concise, and visually unambiguous text prompts that test specific temporal relations: "before", "after", "until", and "then".
GENERAL CONSTRAINTS FOR ALL PROMPTS :

- Short Timeframe: T2V models typically generate 5-8 seconds of video. The events must happen in quick, smooth succession and easily fit within this brief window.
- Strictly Visual: All actions must be 100% visually identifiable. Do NOT include any sound-related cues (e.g., speaking, alarm ringing).
- Coverage: Include humans, animals, daily objects, and common actions and events.
- High Visual Contrast: The two events/actions must be highly distinct and visually noticeable so human evaluators can easily determine their temporal order.
- Concise Language: Write clear, single-sentence prompts. Avoid overly complex descriptions.

Input:
Temporal relation: "before" / "after"
Rule: Reversible pairs evaluating logical order. Every prompt must be generated as a pair: [A before B] and [A after B].
Actor Categories:
- One-Actor: One subject performs two reversible actions.
- Two-Actor: Two subjects perform reversible actions.

Please generate 50 prompts in total, with 25 per action category.

Input:
Temporal relation: "until"
Rule: A continuous action stops exactly when a terminating event begins.
Actor Categories:
- One-Actor: A single subject undergoes a continuous process until it reaches a final state.
- Two-Actor: Subject 1 performs a continuous action that stops exactly when it triggers an event in Subject 2.

Please generate 50 prompts in total, with 25 per action category.

Input:
Temporal relation: "then"
Rule: Strict sequential order. The first action finishes (or establishes a lasting visual state) before the second action starts.
Actor Categories:
- One-Actor: One subject does A, then the same subject does B.
- Two-Actor Subject 1 does A, then Subject 2 does B in the same scene.

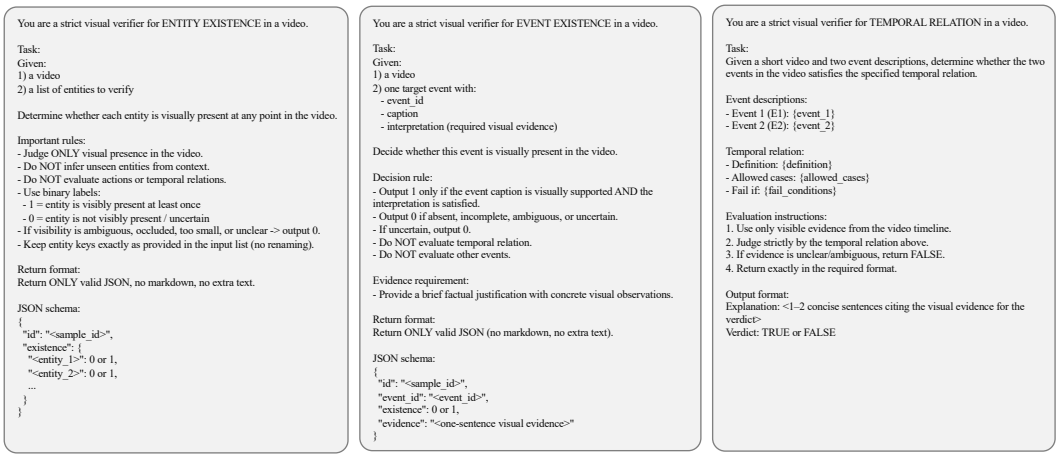
Please generate 50 prompts in total, with 25 per action category.

Figure 6: Template for text prompt generation.

in Figure 7 (a). Let v_i denote the i-th generated video and $\mathcal{O}^{(i)}$ denote the set of required entities extracted from the prompt. For each entity $o \in \mathcal{O}^{(i)}$, the MLLM is asked to determine whether the entity appears in the video based purely on visual evidence. The entity existence score is defined as:

$$S_{ent} = \frac{1}{N} \sum_{i=1}^{N} \frac{1}{|\mathcal{O}^{(i)}|} \sum_{o \in \mathcal{O}^{(i)}} \mathbb{I}(o \text{ is present in } v_i) \tag{13}$$

This metric captures the average grounding success across all entities and prompts.



(a) Entity Existence (b) Event Existence (c) Temporal Relation

Figure 7: Template for MLLM-based evaluation.

Table 7: Canonical temporal operators and illustrative examples.

Operator	**Example**
before	A squirrel picks up a pine cone *before* climbing onto a branch. A person opens an umbrella *before* stepping into the rain. A frog jumps onto a lily pad *before* a bird flies away.
after	An owl blinks *after* turning its head. A person puts a cookie on a plate *after* pouring milk into a glass. A ball rolls into frame *after* a toy block falls over.
until	A spider lowers itself on a silk thread *until* it reaches the ground. A pendulum swings *until* a person catches it. A person scrubs a dirty dish *until* the stains disappear.
then	A person stamps an envelope, *then* puts it into a basket. A cat knocks over a glass, *then* a dog runs away. A bird lands on its nest, *then* a chick opens its beak.

Event Existence. This dimension evaluates whether the required dynamic events occur in the generated video. To reduce ambiguity in event interpretation, each event is paired with a short interpretation (visual clue) describing observable motion or interaction. For example:

- Event description: “a person picks up the phone ”
- Interpretation: “visible grasping and lifting motion of the phone.”

A video depicting only the final state (e.g., a person already holding the phone) would therefore be considered a failure. The prompt template used for event existence is provided in Figure 7 (b). For prompts containing two events, P and Q, the event existence score is computed as:

$$S_{evt} = \frac{1}{N}\sum_{i=1}^{N}\frac{\mathbb{I}(P^{(i)} \text{ occurs in } v_i) + \mathbb{I}(Q^{(i)} \text{ occurs in } v_i)}{2} \tag{14}$$

This stage ensures that temporal evaluation is performed only when the required actions are actually realized.

Temporal Relation. Temporal relations are evaluated only when both events are successfully detected, i.e., $S_{evt}^{(i)} = 1$. This dependency prevents incorrect temporal conclusions caused by missing events.

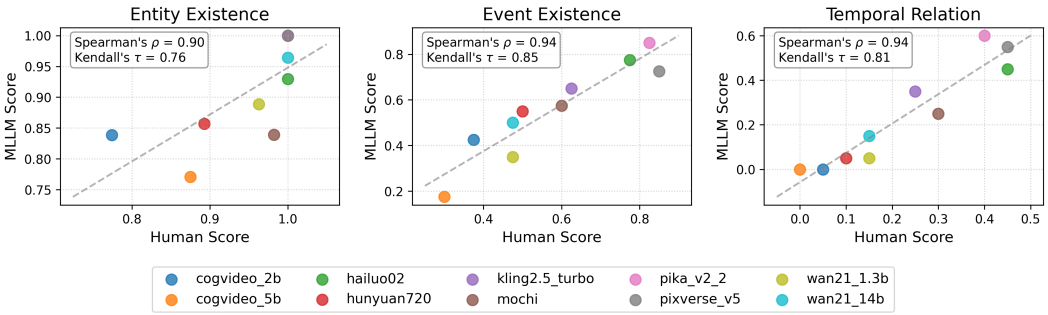


Figure 8: Validate TempBench's Human Alignment. Each point represents a model, comparing human- and MLLM-computed satisfaction scores for entity existence, event existence, and temporal relation. The strong Spearman and Kendall correlations indicate that the MLLM-based verifier reliably preserves human-assessed model trends across all three evaluation dimensions.

For each operator, the MLLM receives: (i) the event descriptions, (ii) a strict definition of the temporal relation, (iii) allowed cases, and (iv) explicit failure conditions. The model then returns a binary decision indicating whether the relation is satisfied. The prompt template and criteria for temporal relation are provided in Figure 7 (c) and Figure 14, respectively. The overall temporal relation score is calculated as:

$$S_{rel} = \frac{1}{N}\sum_{i=1}^{N}\mathbb{I}\Big(S_{evt}^{(i)} = 1 \wedge R^{(i)} \text{ is satisfied in } v_i\Big). \tag{15}$$

This metric measures strict compliance with the temporal relation specified by the prompt. Overall, these metrics together provide a fine-grained diagnostic view of model performance by separating grounding failures, action failures, and temporal reasoning failures.

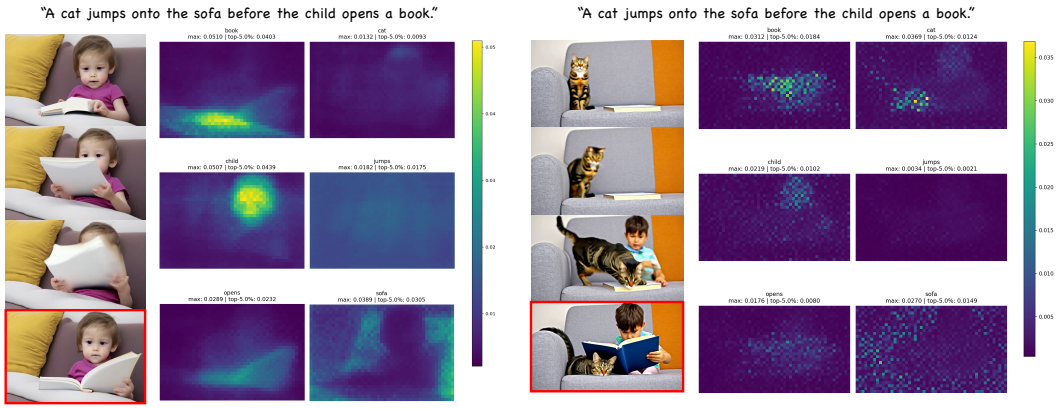
A.3 Evaluation Validation

To validate the reliability of the proposed MLLM-based verifier, we conduct a human evaluation study on a subset of generated videos. We sample videos from all evaluated models to ensure broad coverage. For each model, we randomly select 20 generated videos, including five prompts for each temporal relation. The sampled prompts are unique across models. This procedure yields 200 annotated videos in total, covering all temporal operators and a diverse set of models.

Figure 15 illustrates the annotation interface. Specifically, we manually evaluate each video following the same hierarchical criteria used in TempBench, including entity existence, event existence, and temporal relation satisfaction. To assist manual inspection, we additionally record approximate timestamps for event occurrences as supporting evidence. However, since MLLMs are less reliable at precise temporal localization, especially for short generated clips, we do not use timestamp information in the validation. Instead, we compare only the final binary decisions on the three evaluation dimensions.

Using the human annotations, we recompute S_{ent}, S_{evt}, and S_{rel} for each model on the annotated subset and compare them with the scores produced by the MLLM-based verifier. Figure 8 plots the agreement between human and MLLM evaluations. Each point corresponds to a model, where the x-axis represents the human satisfaction score and the y-axis represents the MLLM score. We report Spearman's ρ and Kendall's τ to measure correlation. Across all three metrics, the MLLM evaluator shows strong alignment with human judgments. These results indicate that the automatic verifier reliably preserves model-level performance trends across entity grounding, event realization, and temporal relation satisfaction.

Table 8 reports agreement at the level of individual decisions on the same 200 annotated videos. Per-video agreement is the fraction of binary labels matching the human annotations within each video, averaged over videos; classification accuracy pools all binary decisions within each evaluation dimension. The two measures are close across all three dimensions, with the highest agreement for entity existence.



CogVideoX-2B Wan 2.1-1.3B

Figure 9: Visualization of token-level attention activations. Each heatmap shows frame-level spatial attention for representative entity and action tokens, forming a natural interface between text tokens and visual tokens that enables grounding verification of entities and actions.

Table 8: Instance-level agreement between the MLLM verifier and human annotations on TempBench.

Dimension	Per-video agreement (%)	Classification accuracy (%)
Entity existence	93.40	93.27
Event realization	75.25	75.25
Temporal relation	80.50	80.50

B More Implementation Details

Clean Latent Estimation and Re-noising. Before performing latent perturbation and proposal search, LogicDirector first estimates the corresponding clean latent $\hat{\mathbf{z}}_0$ from the current noisy latent $\mathbf{z}_t$. Since different diffusion backbones adopt different training paradigms, the inversion step follows the standard formulation associated with each model. Specifically, CogVideoX-2B follows a conventional noise-prediction diffusion formulation, where the network predicts the noise component $\epsilon_\theta(\mathbf{z}_t, \mathbf{c}, t)$ at timestep t. Given the cumulative noise schedule $\bar{\alpha}_t$, the clean latent can be estimated through the standard inversion:

$$\hat{\mathbf{z}}_0 = \frac{\mathbf{z}_t - \sqrt{1-\bar{\alpha}_t}\,\epsilon_\theta(\mathbf{z}_t, \mathbf{c}, t)}{\sqrt{\bar{\alpha}_t}}. \tag{16}$$

This equation directly follows from the forward diffusion formulation $\mathbf{z}_t = \sqrt{\bar{\alpha}_t}\mathbf{z}_0 + \sqrt{1-\bar{\alpha}_t}\boldsymbol{\epsilon}$, allowing the model to reconstruct an estimate of the clean latent $\mathbf{z}_0$ from the predicted noise.

In contrast, Wan 2.1-1.3B is trained under a flow-matching paradigm, where the trajectory is parameterized as a linear interpolation between clean latents and Gaussian noise. Let σ_t denote the noise scale at timestep t, obtained from the scheduler. The predicted velocity v_θ represents the direction from the clean latent toward the noisy state. Therefore, the clean latent estimate can be directly computed as:

$$\hat{\mathbf{z}}_0 = \mathbf{z}_t - \sigma_t\, v_\theta(\mathbf{z}_t, \mathbf{c}, t). \tag{17}$$

This formulation follows the linear interpolation structure of flow-matching models, where the current latent $\mathbf{z}_t$ is expressed as the clean latent perturbed along a noise direction scaled by σ_t. Subtracting the predicted velocity therefore yields a consistent estimate of the clean latent. In both cases, the estimated $\hat{\mathbf{z}}_0$ serves as the reference latent for subsequent perturbation and proposal generation in the Best-of-n search procedure. For the re-noising step in Eq. (11), the coefficients follow each backbone's forward noising process: $(a_t, b_t) = (\sqrt{\bar{\alpha}_t}, \sqrt{1-\bar{\alpha}_t})$ for CogVideoX-2B and $(a_t, b_t) = (1-\sigma_t, \sigma_t)$ for Wan 2.1-1.3B.

Attention Map Extraction. CogVideoX [60] adopts a Multi-Modal Diffusion Transformer (MMDiT) with global self-attention over concatenated video-text tokens. To apply LogicDirector, we extract the

text-to-video attention sub-matrices from all 30 MMDiT blocks and aggregate them into spatiotemporal attention maps. For Wan 2.1 [52], which employs explicit cross-attention layers, we directly collect cross-attention maps from all DiT blocks.

Attention Activation Threshold. For logical constraint computation, we introduce an attention activation threshold γ to identify salient regions in token-level attention maps. We set γ to 0.04 for entities and 0.025 for actions in CogVideoX-2B, and to 0.025 for both entities and actions in Wan 2.1-1.3B. These values are chosen empirically based on visualizations of the two models' attention maps, as shown in Figure 9. We find that the two architectures exhibit distinct activation patterns. In CogVideoX-2B, video-text submatrices extracted from self-attention maps yield relatively dense activations: visible entities typically correspond to values around 0.004, while action tokens reach higher responses around 0.02–0.03. By contrast, Wan 2.1-1.3B produces sparser cross-attention maps, with visible entities typically activating around 0.02–0.03 and actions around 0.02. We observe similar activation patterns across different prompts and generated videos, indicating that these thresholds provide stable detection of entities and actions while balancing sensitivity to foreground evidence against robustness to noisy activations.

Loss Weight Configurations. The three logical constraint terms operate on different numerical scales. In practice, the existence loss typically falls in the range of 10^{-4}–10^{-2}, the binding loss is around 10^{-1}, and the temporal relation loss is around 2–3. To balance their contributions, we adopt a normalized base configuration with $\lambda_1 = 1.0$, $\lambda_2 = 0.1$, and $\lambda_3 = 0.005$ for $\mathcal{L}_{exist}$, $\mathcal{L}_{bind}$, and $\mathcal{L}_{rel}$, respectively. We then apply minor relation- and backbone-specific adjustments, guided by the empirical observations in Table 3. For CogVideoX-2B, *before* and *then* already achieve moderate existence satisfaction under the base configuration. However, *then* requires clearer separation between the two events, so we increase the temporal relation weight λ_3 to 0.05. In contrast, *after* and *until* still struggle with entity and event existence; accordingly, we increase λ_1 to 5.0 to encourage stronger grounding of the required entities and actions. The action-binding weight λ_2 remains fixed at 0.1 across all relations. For Wan2.1-1.3B, spatial grounding is generally more stable, allowing λ_1 to remain fixed at 1.0 for all operators. The *after* and *until* relations work well under the base configuration. Similar to CogVideoX-2B, *then* benefits from stronger temporal separation, so we increase λ_3 to 0.05. For *before*, we apply a minor adjustment with $\lambda_2 = 0.05$ and $\lambda_3 = 0.01$. Overall, these bounded adjustments mainly compensate for differences in backbone architectures and operator difficulty while preserving the robustness of the core objective design throughout our experiments.

C Qualitative Analysis of Best-of-*n* Latent Search

To further examine the behavior of Best-of-*n* latent search, we visualize a representative example showing how the proposal count n, the inner-loop iterations K, and the guidance steps T affect generation. As shown in Figure 10, the baseline model correctly renders the two actions *jump* and *flies away*, but merges them into a concurrent event instead of preserving the intended temporal order. After applying LogicDirector, the outputs show consistently better temporal composition across different search configurations. In addition, we observe that:

(1) Minimal search already provides improvements. The configuration $(1, 1, 10)$ already improves over the vanilla output, indicating that LogicDirector does not require a large proposal pool to be effective. Here, n counts only the *perturbed* proposals. Thus, even when $n = 1$, the method still performs a non-trivial search by comparing the single perturbed candidate with the original latent z_t and selecting the better one. This suggests that even one refined perturbation is often sufficient to correct an unfavorable generation trajectory.

(2) Proposal refinement is more influential than proposal diversity in the low-search regime. Although $(3, 1, 10)$ introduces broader stochastic exploration through multiple perturbed candidates, it still fails to clearly render the first event. In contrast, $(1, 3, 10)$ uses only a single perturbed proposal yet produces a clearer realization of both sub-events. This indicates that improving the quality of each proposal through iterative refinement is more important than simply increasing proposal diversity.

(3) Excessive proposal diversity can introduce unstable candidates. Expanding the proposal pool to $n = 3$ generates multiple independently perturbed candidates that are compared against the original latent z_t. While broader exploration can sometimes improve temporal ordering, it also increases the likelihood of selecting visually unstable trajectories. For example, under the $(3, 3, 10)$ configuration,

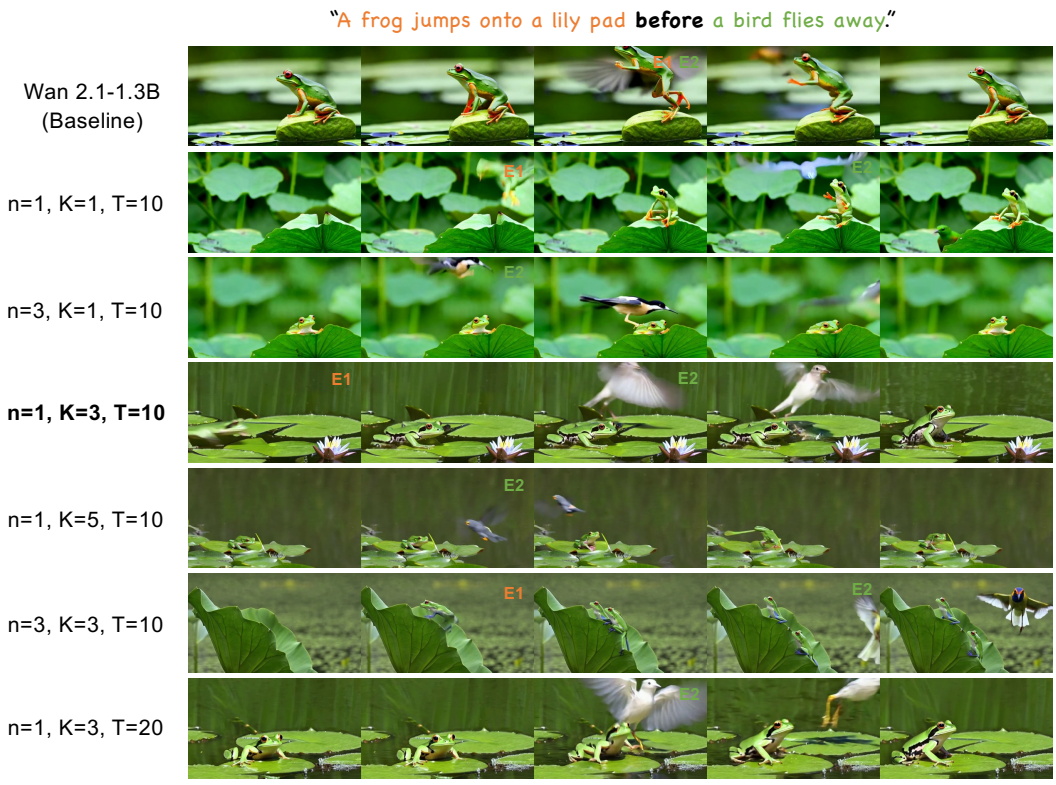


Figure 10: Effect of different setups in Best-of-n latent search.

the selected candidate exhibits noticeable structural artifacts (e.g., one frog becomes two), suggesting that overly permissive exploration may trade visual fidelity for improved heuristic scores.

(4) Stronger search is not always beneficial. Increasing K too aggressively may over-refine a proposal and weaken one of the sub-events. Similarly, extending the guidance window from $T = 10$ to $T = 20$ does not further improve performance and may even degrade visual fidelity. This suggests that logical guidance is most effective when applied with moderate strength during early denoising steps rather than over an extended trajectory.

Overall, the three hyperparameters play distinct roles: n controls proposal diversity, K controls the refinement depth of each proposal, and T determines how long the denoising trajectory is actively steered. These results suggest that improving temporal composition requires a balanced search strategy, in which moderate refinement and controlled exploration yield the most reliable outcomes.

D Comparison with other test-time scaling methods

We compare LogicDirector with two representative test-time scaling approaches: Best-of-n seed search and prompt enhancement. For Best-of-n, we generate multiple samples with different random seeds and select the best using the ViCLIP [54] score. For prompt enhancement, we use an LLM [38] to expand the prompt with richer context while preserving the intended temporal order. As shown in Table 9, Best-of-n seed search yields only modest improvements in temporal correctness while increasing runtime roughly linearly with n, suggesting that generic similarity-based reranking is inefficient for enforcing event ordering. Prompt enhancement is a strong practical baseline, improving event realization and temporal cor-

Table 9: Performance comparison with different test-time scaling methods.

Models	S_{ent}	S_{evt}	S_{rel}	Time (s)
Wan 2.1-1.3B	87.2	46.5	13.0	118
+ Seed Search ($n = 2$)	90.2	44.8	16.0	236
+ Seed Search ($n = 4$)	92.2	48.3	17.5	472
+ Prompt Enhancement	88.6	53.0	24.0	124
+ **LogicDirector (Ours)**	90.2	53.0	24.5	169
+ PE + LogicDirector	90.1	57.5	27.5	174



Figure 11: Qualitative examples of two challenging cases from StoryEval.

rectness with little overhead. However, it changes the input specification by expanding an abstract temporal relation into an explicit narrative, thereby externalizing part of the temporal reasoning process before generation. In this sense, it partially bypasses the intrinsic temporal-composition challenge: the T2V model is asked to render a more resolved script rather than infer temporal structure from the original prompt. This may also partly explain the strong performance of commercial systems, which often rely on proprietary prompt rewriting, planning, or prompt optimization pipelines. Notably, combining an enhanced prompt with LogicDirector further improves both event realization and temporal relations. This indicates that the two strategies are complementary: prompt enhancement improves the input specification, whereas LogicDirector guides temporal compositionality during denoising.

E Additional Qualitative Comparisons

Figures 16 and 17 provide additional qualitative comparisons between the baseline models and LogicDirector on TempBench. These examples show that LogicDirector consistently enhances event realization and temporal ordering across different backbones and operators, supporting our quantitative findings that logic-guided search steers diffusion trajectories toward more temporally consistent generations.

F Generalization to Other Benchmarks

T2V-CompBench. To further validate the generalization capability of LogicDirector, we conduct additional experiments on two dimensions of T2V-CompBench [47], namely *Action Binding* and *Object Interaction*. The *Action Binding* dimension evaluates whether a model can correctly associate multiple actions with their corresponding objects when both are described in the prompt. The *Object Interaction* dimension tests whether a model can understand and generate dynamic interactions, including physical interactions that cause motion or state changes, as well as social interactions between animate entities. For these experiments, we apply LogicDirector using only the existence and action-binding constraints, while keeping the remaining settings the same as in TempBench. As shown in Table 10, LogicDirector consistently improves the performance of Wan 2.1-1.3B. The gain is more noticeable on *Object Interaction*, where successful generation requires both correct grounding and coherent interaction modeling. We also observe a smaller but consistent improvement on *Action Binding*, suggesting that the proposed existence and binding constraints remain beneficial beyond the temporal setting. Overall, these results indicate that LogicDirector can generalize as a lightweight test-time guidance

Table 10: Evaluation on T2V-CompBench.

Models	Action Binding	Object Interaction
Wan2.1-1.3B	0.6546	0.6281
+ LogicDirector (Ours)	0.6569	0.6467

strategy for improving compositional faithfulness in broader text-to-video generation tasks. More results can be found in Figure 18 and Figure 19.

TC-Bench. TC-Bench [14] contains 150 prompts describing temporal evolution through attribute transitions, object-relation changes, and background shifts without necessarily using an explicit temporal connective. For these prompts, the GPT-4.1 parser maps the described progression to the closest supported temporal operator before LogicDirector is applied. Table 11 reports Temporal Consistency Rate (TCR) and TC-Score for Wan 2.1-1.3B. LogicDirector improves TCR in all three categories and raises overall TCR from 8.667 to 14.000. Overall TC-Score rises from 0.6750 to 0.6845, while the background TC-Score decreases slightly.

Table 11: Results on TC-Bench. TCR is reported in percent; higher is better for both metrics.

	Attribute		Object		Background		Overall	
Method	TCR	TC-Score	TCR	TC-Score	TCR	TC-Score	TCR	TC-Score
Wan 2.1-1.3B	7.692	0.6708	12.766	0.7319	5.882	0.6268	8.667	0.6750
+ LogicDirector	17.308	0.7026	14.894	0.7362	9.804	0.6184	14.000	0.6845

StoryEval. To probe the boundaries of our framework, we qualitatively evaluate LogicDirector on more complex multi-event prompts sampled from the challenging StoryEval [55] benchmark. As shown in Figure 11, LogicDirector exhibits encouraging generalization beyond the two-event setting. In the first example, our method produces a more recognizable three-stage progression, whereas the baseline Wan 2.1-1.3B model largely collapses the sequence into a dominant *licking* action, with weak or missing boundaries for *opening* and *closing* the jar. In the spider prompt, LogicDirector also improves the rendering of both the concurrent and sequential structure, although the full composition remains only partially satisfied: the final directional reversal (*moving left*) is still incomplete. We hypothesize that this limitation mainly stems from the restricted temporal capacity of the underlying video generator rather than the search procedure itself. When three or more distinct events are compressed into a short video generation window, they must compete for the same finite temporal budget. As a result, although LogicDirector can better organize the latent trajectory toward the desired logic, the limited video length still constrains the full realization of long-horizon narratives.

G Failure Analysis and Discussion

Although LogicDirector consistently improves temporal composition, several recurring failure modes remain. **First**, latent search may exploit shortcuts in the existence objective by transforming one object into another instead of preserving stable object identity. As shown in Figure 12, the bee suddenly becomes a butterfly in the first video. This is the most common failure we observe. One possible reason is that the current existence constraint checks whether evidence of the required object appears in the attention maps, but does not explicitly enforce temporal identity consistency. As a result, the search may favor trajectories that satisfy object presence while violating object permanence. **Second**, the model may render an event as a *final state* rather than a transition. For example, the second and third videos show the completed states of "lands on" and "stepping onto the podium," instead of clearly depicting the instantaneous action itself. This likely stems from the limited temporal dynamics modeling of current video generators, which often compress actions into their final visual outcomes. In such cases, LogicDirector can help recover event ordering, but cannot fully force the backbone to synthesize missing transitions. **Third**, although our method improves event ordering, it does not explicitly reason about causal consequences. In the last video, the two events occur in the required order, but the campfire does not go out afterward. This suggests that our operator-level constraints are effective for



Figure 12: Visualization of failure cases.

temporal ordering but remain limited in modeling deeper causal or state-change reasoning. **Finally**, LogicDirector performs only a finite search within a restricted denoising window, so some failures cannot be corrected through latent search alone. A promising direction is to extend the current logic verifier with identity-preserving, transition-aware, and causal state-change constraints, possibly by incorporating stronger spatiotemporal grounding signals beyond attention alone.

System-Msg: "You are an expert linguistic parser. Your task is to extract variables and predicates from text-to-video prompts.
- Variables: The distinct, visible entities (humans, animals, objects) depicted in the scene.
- Predicates: The actions, attributes, or temporal relations connecting those entities.
TASK:
Given ONE sentence (the 'Prompt'), output a JSON object with a 'propositions' field containing:
1) 'Existence' (Variables): list all distinct VISIBLE entities mentioned in the prompt.
2) 'Implication' (Predicates): map each ACTION or ATTRIBUTE to its related entities (subject and/or object) using the format: \"<ACTION>\": [\"<subject>\", \"<object>\"]
- ACTION must be the core verb phrase. This can be a single verb ('scratches'), a verb+particle ('picking up'), or a light-verb construction ('taking a bow', 'takes a large bite').
- If the ACTION is a light-verb construction, the noun ('bite') is part of the action and should NOT be listed as a separate object or in the 'Existence' list.
- If an action has no object, list only the subject.
3) 'Temporal_Relation' (Predicates): A dictionary with exactly ONE key chosen from ['before', 'after', 'until', 'then'], mapping to a list of two actions: [\"<ACTION_1>\", \"<ACTION_2>\"]. Both elements MUST exactly match keys in 'Implication'.
RULES:
- VERBATIM SPANS (NO ARTICLES): All text for entities and actions must be copied exactly from the prompt, preserving original capitalization and tenses (DO NOT lemmatize). However, you MUST remove leading articles/determiners ('the', 'a', 'an').
- RESOLVE PRONOUNS: You MUST resolve pronouns (like 'it', 'he', 'she', 'they') to their antecedent noun phrase. In the 'Implication' map, use the verbatim text span of the *antecedent noun* (without articles), not the pronoun. For example, in 'A balloon inflates until it pops', the 'Implication' must be: `\"pops\": [\"balloon\"]`.
- IGNORE META-ENTITIES: Do not list words that refer to the video container, boundary, or camera. Only list physical entities inside the scene. (e.g., in '...rolls into frame', 'frame' is excluded).
- NULL HANDLING: If the prompt contains no actions, set 'Implication' to null. If there is no temporal sequence, set 'Temporal_Relation' to null.
- Return ONLY valid JSON (no explanations or code fences).
OUTPUT FORMAT:

```
{\n
 \"prompt\": \"<original prompt>\",\n
 \"propositions\": {\n
  \"Existence\": [\"<entity_a>\", \"<entity_b>\", \"<entity_c>\"],\n
  \"Implication\": {\n"
     \"<action_1>\": [\"<entity_a>\", \"<entity_b>\"],\n
     \"<action_2>\": [\"<entity_c>\"]\n
   },\n
   \"Temporal_Relation\": {\n
     \"<temporal_keyword>\": [\"<action_1>\", \"<action_2>\"]\n
   } \n
  } \n
 } \n
```

Figure 13: Template for LLM-based text prompt parsing.

```
{
  "before": {
    "definition": "Event 1 must start visibly earlier in the video than Event 2.",
    "allowed": [
      "E1 and E2 may overlap later in the video, as long as E1 clearly initiates first."
    ],
    "fail_conditions": [
      "Event 2 starts earlier than Event 1.",
      "Event 1 and Event 2 start at the exact same time."
    ]
  },
  "after": {
    "definition": "Event 2 must start visibly earlier in the video than Event 1.",
    "allowed": [
      "E1 and E2 may overlap later in the video, as long as E2 clearly initiates first."
    ],
    "fail_conditions": [
      "Event 1 starts earlier than Event 2.",
      "Event 1 and Event 2 start at the exact same time."
    ]
  },
  "then": {
    "definition": "Event 1 must be completely finished before Event 2 starts.",
    "allowed": [
      "A brief, natural temporal pause or gap between the end of E1 and the start of E2 is acceptable."
    ],
    "fail_conditions": [
      "Event 2 starts while Event 1 is still happening (any temporal overlap).",
      "Event 2 happens before Event 1."
    ]
  },
  "until": {
    "definition": "Event 1 is a continuous state or action that must hold up to the first occurrence of Event 2,
which interrupts or terminates Event 1.",
    "allowed": [
      "None."
    ],
    "fail_conditions": [
      "Event 1 stops before Event 2 begins (a temporal gap).",
      "Event 1 continues unchanged long after Event 2 has occurred."
    ]
  }
}
```


Figure 14: Criteria for different temporal relations.

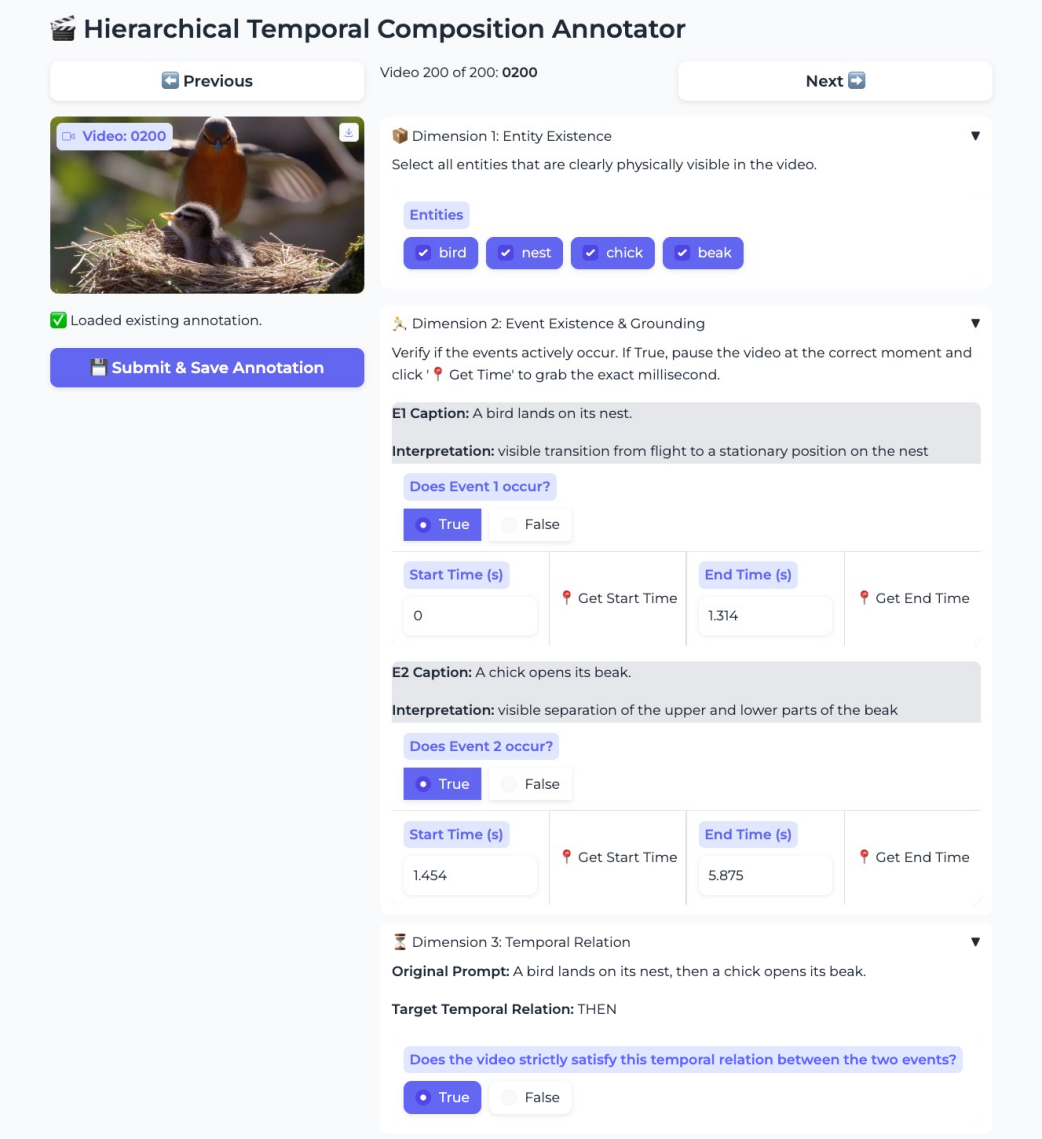


Figure 15: Web interface of human annotation.

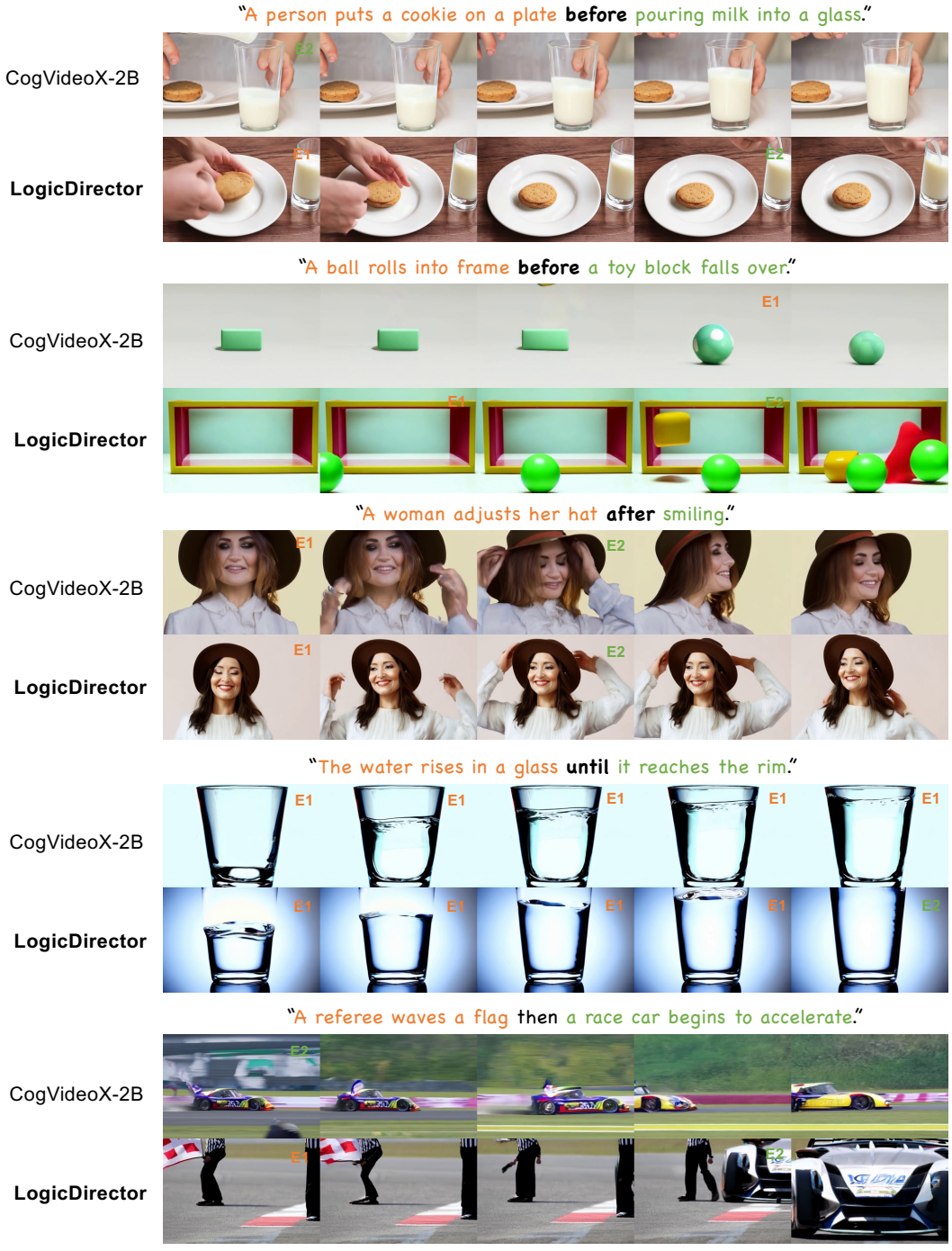


Figure 16: More qualitative comparison between CogVideoX-2B and LogicDirector.

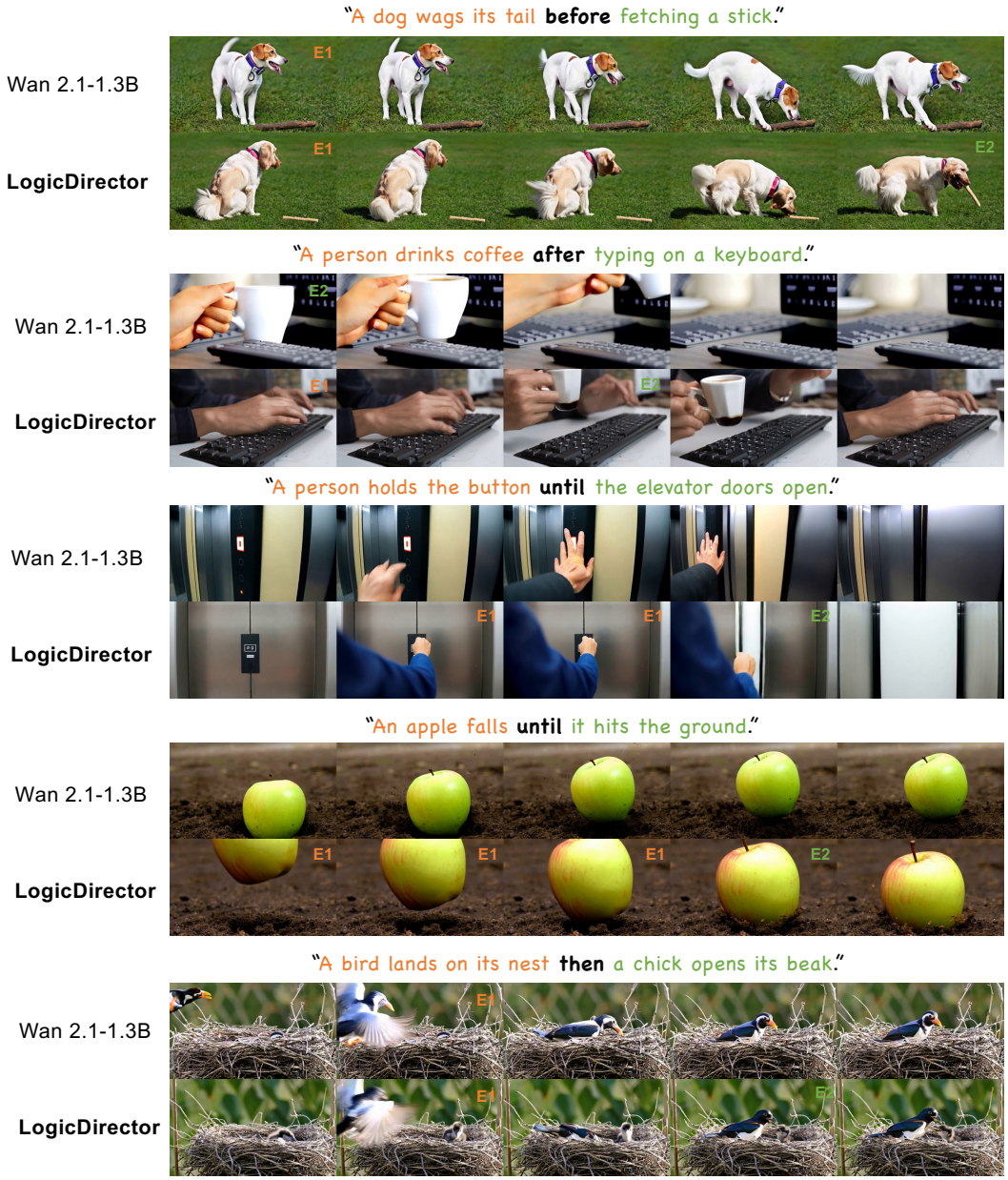


Figure 17: More qualitative comparison between Wan 2.1-1.3B and LogicDirector.

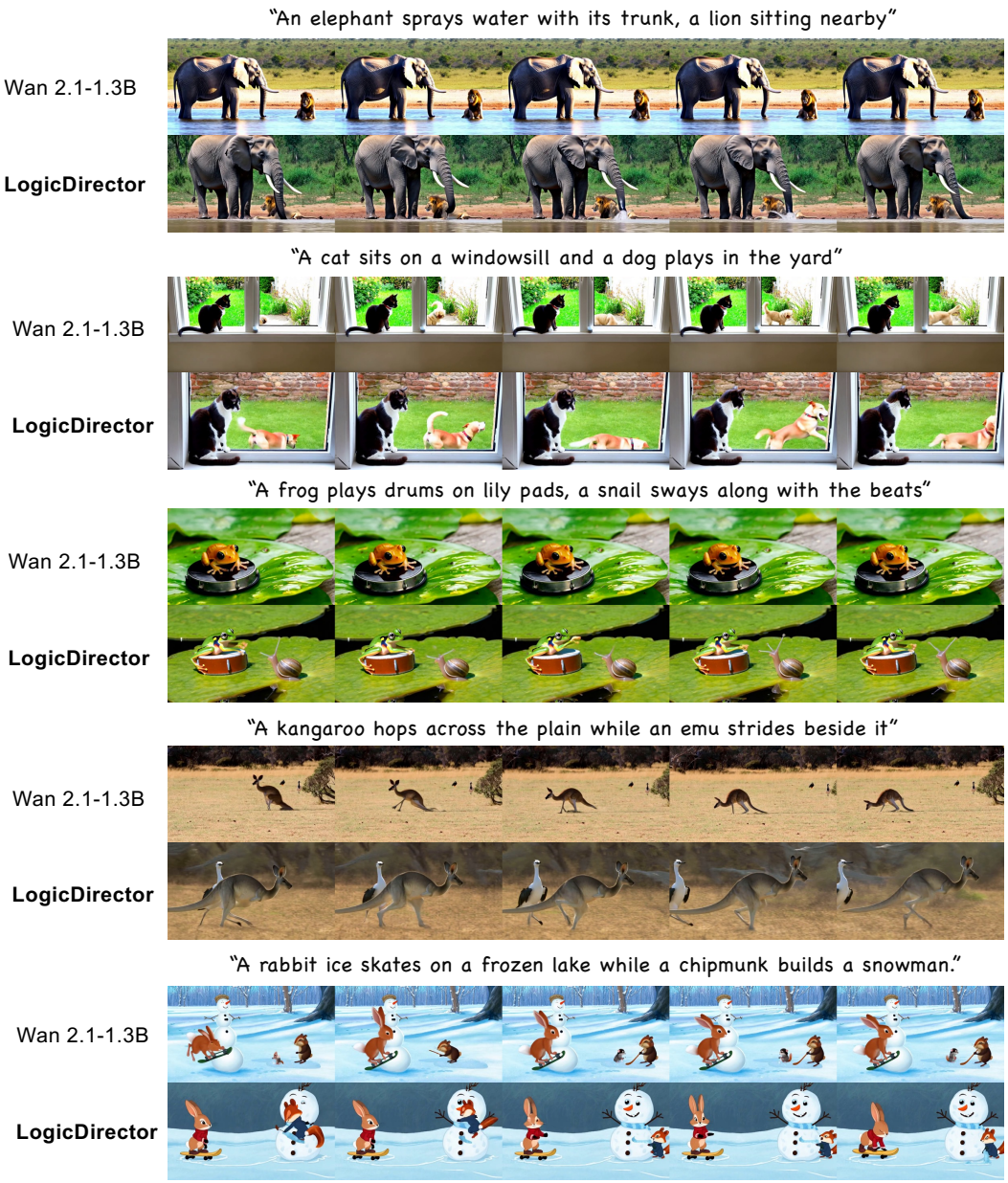


Figure 18: Visualizations of *Action Binding* on T2V-CompBench.

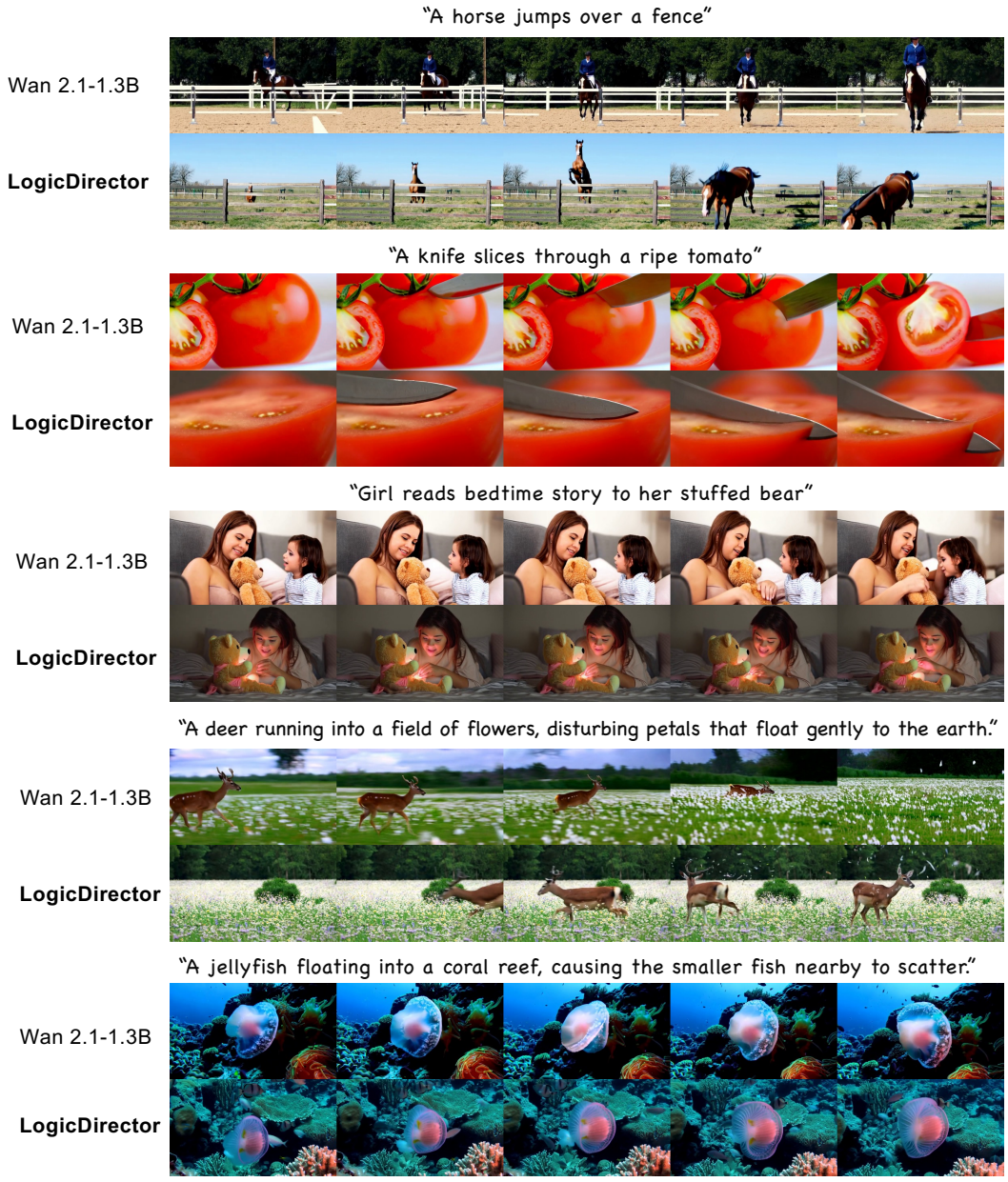


Figure 19: Visualizations of *Object Interaction* on T2V-CompBench.

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